

CPEN 416 / EECE 5710

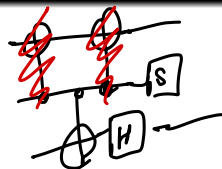
Lecture 12: The quantum Fourier transform (QFT)

Tuesday 21 October 2025

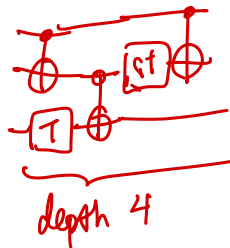
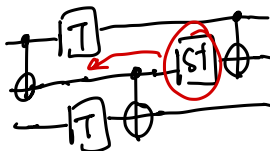
Announcements



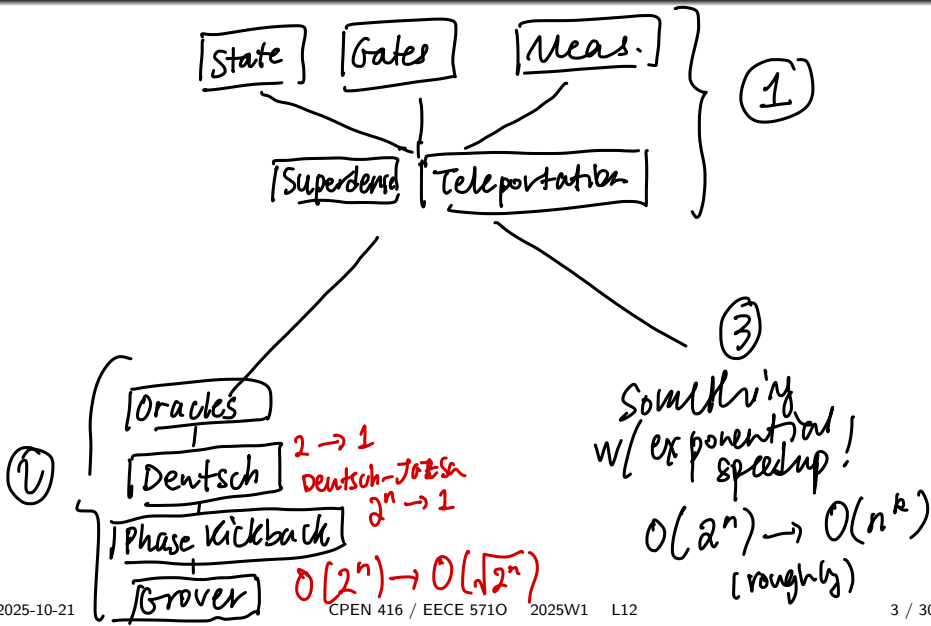
fourier



- Quiz 5 today
- Tutorial Assignment 2 due Friday at 23:59
- Final exam scheduled Thursday 11 December 15:30-18:00



Where have we been?



Where are we going?

```
def shors_algorithm(N):
    p, q = 0, 0

    while p * q != N:
        a = np.random.choice(list(range(2, N - 1)))

        if np.gcd(a, N) != 1:
            p = np.gcd(a, N)
            q = N // p
            return p, q

        sample = get_sample(a, N)
        phase = fractional_binary_to_float(sample)
        candidate_order = phase_to_order(phase, N)

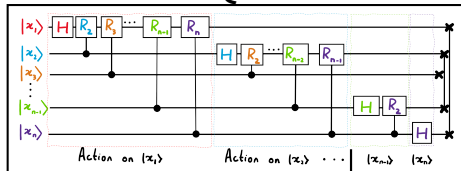
        if candidate_order % 2 == 0:
            square_root = (a ** (candidate_order // 2)) % N

            if square_root not in [1, N - 1]:
                p = np.gcd(square_root - 1, N)
                q = np.gcd(square_root + 1, N)

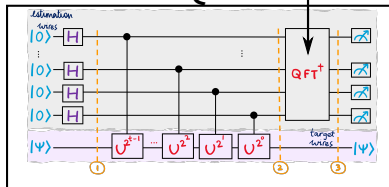
    return p, q
```

Where are we going?

1. QFT



2. QPE



3. Shor

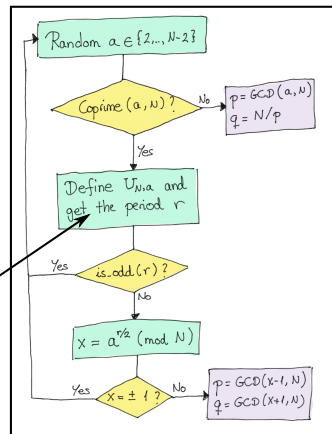


Image credits: PennyLane Codebook

Module 3 learning outcomes

Learning outcomes:

- define, and state the scaling of, the quantum Fourier transform
- use quantum phase estimation to determine the eigenvalues of a unitary matrix
- use the QFT and QPE as subroutines to implement order finding, and simulate Shor's factoring algorithm
- identify cryptographic schemes susceptible to quantum attack
- describe the societal and ethical implications of quantum technology

Today

Learning outcomes:

- Express floating-point values in fractional binary representation
- Describe the behaviour of the quantum Fourier transform
- Construct a circuit for the quantum Fourier transform and analyze its resource usage

Fractional binary

Consider binary representation of an integer k ,

$$k = 2^{n-1} \cdot k_1 + 2^{n-2} k_2 + \dots + 2 k_{n-1} + k_n$$
$$k_j \in \{0, 1\}$$

We can extend this notation to floating point numbers:

$$0.k_1 k_2 k_3 \dots k_n = 2^{-1} k_1 + 2^{-2} k_2 + \dots + 2^{-n} k_n$$
$$= \frac{k_1}{2} + \frac{k_2}{4} + \dots + \frac{k_n}{2^n}$$

$$\frac{1}{4} = 0.01$$

Fractional binary

Exercise: let $k = 0.11010$. What is the numerical value of k ?

$$\begin{aligned} 0.11010 &= 1 \cdot \frac{1}{2} + 1 \cdot \frac{1}{2^2} + 0 \cdot \frac{1}{2^3} + 1 \cdot \frac{1}{2^4} \\ &= \frac{1}{2} + \frac{1}{4} + \frac{1}{16} \\ &= 0.8125 \end{aligned}$$

Fractional binary

The QFT will be defined using powers of complex roots of unity:

$$W = e^{\frac{2\pi i}{N}}$$

↓
 $N = 2^n$

$$W^{xk} = W^{\frac{x 2\pi i k}{N}}$$

$$= W^{x 2\pi i \left(\frac{k}{N}\right)}$$

we'll express
in fractional
binary.

Fractional binary representation will help us

- ① derive a circuit implementation for the QFT today
- ② interpret the results of *quantum phase estimation* on Thursday

$$\frac{k}{2^n} = \frac{2^{n-1} k_1}{2^n} + \frac{2^{n-2} k_2}{2^n} + \dots + \frac{2 k_{n-1}}{2^n} + \frac{k_n}{2^n}$$

$$= \frac{k_1}{2} + \frac{k_2}{2^2} + \dots + \frac{k_n}{2^n}$$

The discrete Fourier transform

The DFT and FFT (which implements it efficiently) convert between time and frequency domains in digital signal processing.

$$DFT = \begin{pmatrix} 1 & 1 & 1 & \dots & 1 \\ 1 & \bar{\omega} & \bar{\omega}^2 & \dots & \bar{\omega}^{N-1} \\ 1 & \bar{\omega}^2 & \bar{\omega}^4 & \dots & \bar{\omega}^{2(N-1)} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 1 & \bar{\omega}^{N-1} & \bar{\omega}^{2(N-1)} & \dots & \bar{\omega}^{(N-1)(N-1)} \end{pmatrix}$$

where $\bar{\omega} = e^{-2\pi i/N}$.

The discrete Fourier transform

Given a signal $x[n]$, the DFT computes

$$X[k] = \sum_{n=0}^{N-1} e^{-\frac{2\pi i k n}{N}} x[n] = \sum_{n=0}^{N-1} \bar{\omega}^{nk} x[n]$$

The inverse DFT computes

$$x[n] = \frac{1}{N} \sum_{k=0}^{N-1} e^{\frac{2\pi i k n}{N}} X[k] = \frac{1}{N} \sum_{k=0}^{N-1} \omega^{nk} X[k]$$

where $\omega = e^{2\pi i/N} = \bar{\omega}^{-1}$

QFT (up to prefactor)

Quantum Fourier transform

The quantum Fourier transform (QFT) is the quantum analog of the **inverse DFT**. $|k\rangle = |k_1, k_2, \dots, k_n\rangle$

$$QFT = \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \omega^{jk} \underbrace{|k\rangle \langle j|}_{\text{matrix}}, \quad N=2^n$$

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = a \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} + b \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} + \dots$$

$$= a|0\rangle\langle 0| + b|0\rangle\langle 1| + c|1\rangle\langle 0| + \dots$$

Exercise: Apply the QFT to an n -qubit basis state $|x\rangle$

$$\begin{aligned} QFT |x\rangle &= \frac{1}{\sqrt{N}} \sum_{j=0}^{N-1} \sum_{k=0}^{N-1} \omega^{jk} \underbrace{|k\rangle \langle j|}_{\substack{\text{comp.} \\ \text{basis state.}}} |x\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \end{aligned}$$

$x=j$

Quantum Fourier transform: 1 qubit

$$\text{QFT} |x\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \quad \omega = e^{\frac{2\pi i}{N}}$$

$$n=1$$

$$N=2^n=2$$

$$\text{QFT} |0\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 \omega^{0 \cdot k} |k\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 |k\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$\text{QFT} |1\rangle = \frac{1}{\sqrt{2}} \sum_{k=0}^1 \omega^{1 \cdot k} |k\rangle = \frac{1}{\sqrt{2}} (|0\rangle + \omega |1\rangle) \quad \omega = e^{i\pi} = -1$$

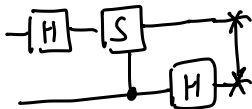


$$\begin{aligned} \text{QFT} |x_1\rangle &= \frac{1}{\sqrt{2}} (|0\rangle + \omega^{x_1} |1\rangle) = \frac{1}{\sqrt{2}} (|0\rangle + e^{\frac{2\pi i x_1}{2}} |1\rangle) \\ &= \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i \cdot 0 \cdot x_1} |1\rangle) \end{aligned}$$

Quantum Fourier transform: 2 qubits

$$QFT = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & \omega^4 & \omega^6 \\ 1 & \omega^3 & \omega^6 & \omega^9 \end{pmatrix} = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & i & -1 & -i \\ 1 & -1 & 1 & -1 \\ 1 & -i & -1 & i \end{pmatrix}$$

Re-express higher powers of $\omega = e^{\frac{2\pi i}{4}}$:



- S gates
- X gates
- 2 Hadamards

Quantum Fourier transform: 2 qubits

$$QFT = \frac{1}{2} \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & \omega & \omega^2 & \omega^3 \\ 1 & \omega^2 & 1 & \omega^2 \\ 1 & \omega^3 & \omega^2 & \omega^1 \end{pmatrix}$$

How does this affect the computational basis states?

$QFT |00\rangle$

$QFT |01\rangle$

$\vdots$

★ try working this out yourself

Quantum Fourier transform: 2 qubits

$$\text{QFT} |x_1\rangle = \frac{1}{\sqrt{2}} (|0\rangle + e^{2\pi i 0 \cdot x_1} |1\rangle)$$

$$\text{QFT} |x_1 x_2\rangle = \frac{1}{2} (|0\rangle + e^{2\pi i 0 \cdot x_2} |1\rangle) \otimes (|0\rangle + e^{2\pi i 0 \cdot x_1 x_2} |1\rangle)$$

Quantum Fourier transform: 3 qubits

$$QFT = \frac{1}{\sqrt{8}} \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & \dots \\ 1 & \omega & \omega^2 & \omega^3 & \omega^4 & \dots \\ 1 & \omega^2 & \omega^4 & \omega^6 & 1 & \dots \\ 1 & \omega^3 & \omega^6 & \omega & \omega^4 & \dots \\ 1 & \omega^4 & 1 & \omega^4 & 1 & \dots \end{pmatrix}$$

$$QFT|x_1x_2x_3\rangle = \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 0.x_3} |1\rangle \right) \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 0.x_2x_3} |1\rangle \right) \otimes \frac{1}{\sqrt{2}} \left(|0\rangle + e^{2\pi i 0.x_1x_2x_3} |1\rangle \right)$$

Quantum Fourier transform: n qubits

We will start here on Thursday.

General expression:

$$\begin{aligned} |x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\ &= \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \cdots (|0\rangle + e^{2\pi i 0 \cdot x_1 \cdots x_n} |1\rangle)}{\sqrt{N}} \end{aligned}$$

Full mathematical derivation at end of slides.

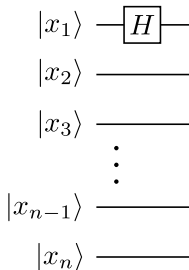
A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0.x_n}|1\rangle)(|0\rangle + e^{2\pi i 0.x_{n-1}x_n}|1\rangle) \cdots (|0\rangle + e^{2\pi i 0.x_1 \cdots x_n}|1\rangle)}{\sqrt{N}}$$

Starting with the state

$$|x\rangle = |x_1 \cdots x_n\rangle,$$

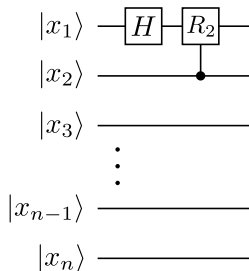
apply a Hadamard to qubit 1:



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0.x_n} |1\rangle) (|0\rangle + e^{2\pi i 0.x_{n-1}x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

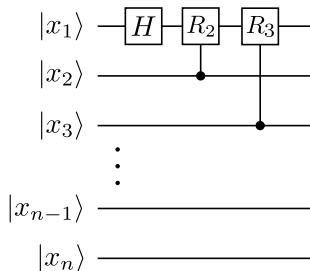
Apply controlled R_2 from qubit
2 $\rightarrow$ 1



A circuit for the QFT

$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0.x_n} |1\rangle) (|0\rangle + e^{2\pi i 0.x_{n-1}x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

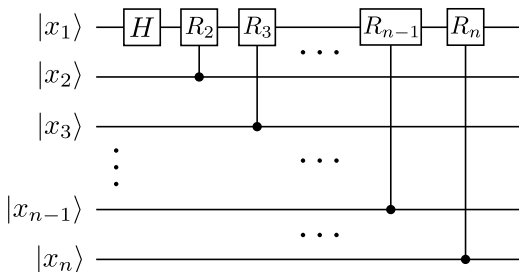
Apply controlled R_3 from qubit
3 $\rightarrow$ 1



A circuit for the QFT

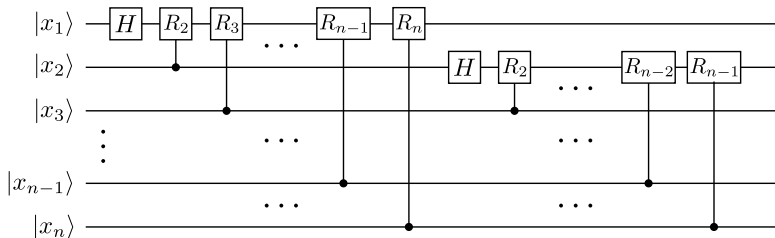
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0 \cdot x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$

Apply a controlled R_4 from $4 \rightarrow 1$, etc. up to the n -th qubit to get



A circuit for the QFT

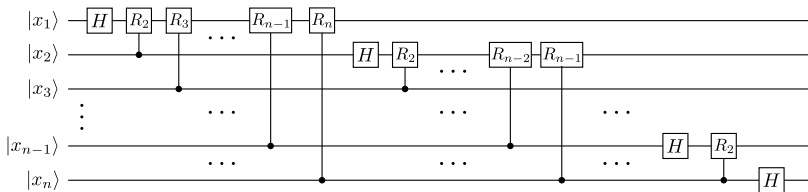
Similar thing with second qubit: apply H then controlled rotations from qubits 3 to n to get



A circuit for the QFT

Do this for all qubits to get that big ugly state from earlier:

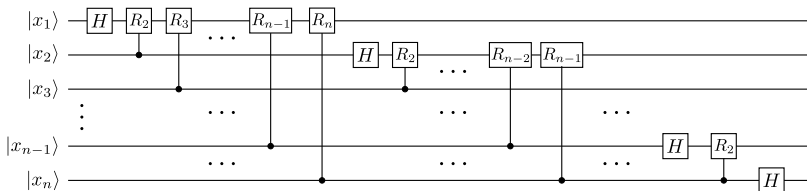
$$|x\rangle \rightarrow \frac{(|0\rangle + e^{2\pi i 0.x_n} |1\rangle) (|0\rangle + e^{2\pi i 0.x_{n-1}x_n} |1\rangle) \dots (|0\rangle + e^{2\pi i 0.x_1 \dots x_n} |1\rangle)}{\sqrt{N}}$$



Qubit order is backwards - easily fixed with SWAP gates.

Quantum Fourier transform

Exercise: What are the gate counts and depth of this circuit?



Next time

Content:

- Quantum phase estimation

Action items:

- ① Tutorial Assignment 2
- ② Work on project

Recommended reading:

- For this class: Codebook module QFT, Nielsen & Chuang 5.1
- For next class: Codebook module QPE, Nielsen & Chuang 5.2

Derivation of output state of QFT

We will reexpress k/N in fractional binary notation, then reshuffle and *factor* the output state.

$$\begin{aligned} |x\rangle &\rightarrow \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} \omega^{xk} |k\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i x (k/N)} |k\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k_1=0}^1 \cdots \sum_{k_n=0}^1 e^{2\pi i x (\sum_{\ell=1}^n k_\ell 2^{-\ell})} |k_1 \cdots k_n\rangle \\ &= \frac{1}{\sqrt{N}} \sum_{k_1=0}^1 \cdots \sum_{k_n=0}^1 \bigotimes_{\ell=1}^n e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \end{aligned}$$

Derivation of output state of QFT

$$\begin{aligned}
 &= \frac{1}{\sqrt{N}} \sum_{k_1=0}^1 \cdots \sum_{k_n=0}^1 \bigotimes_{\ell=1}^n e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \\
 &= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^n \left(\sum_{k_\ell=0}^1 e^{2\pi i x k_\ell 2^{-\ell}} |k_\ell\rangle \right) \\
 &= \frac{1}{\sqrt{N}} \bigotimes_{\ell=1}^n \left(|0\rangle + e^{2\pi i x 2^{-\ell}} |1\rangle \right) \\
 &= \frac{(|0\rangle + e^{2\pi i 0 \cdot x_n} |1\rangle) (|0\rangle + e^{2\pi i 0 \cdot x_{n-1} x_n} |1\rangle) \cdots (|0\rangle + e^{2\pi i 0 \cdot x_1 \cdots x_n} |1\rangle)}{\sqrt{N}}
 \end{aligned}$$